Mapper/TDA Report for Imputed PSCompPars

Se construyeron grafos Mapper sobre el catalogo PSCompPars completado. La imputacion principal es iterative. Los valores se trazan como observed, physically derived o imputed. El objetivo es analisis exploratorio topologico, no prueba final de clases planetarias.

1 Introduction

Este reporte resume un pipeline Mapper/TDA orientado a resultados estaticos. No interpretamos Mapper como prueba directa de la topologia real de los exoplanetas. Interpretamos los grafos como estructuras inducidas por una matriz completada con trazabilidad explicita. Las conclusiones mas confiables son las que persisten bajo cambios de espacio de variables, lens, parametros e imputacion, y que no estan dominadas por variables imputadas o derivadas.

2 Data and Imputation

El dataset usa siete variables principales: `pl_rade`, `pl_bmasse`, `pl_dens`, `pl_orbper`, `pl_orbsmax`, `pl_insol` y `pl_eqt`. Mapper usa una matriz transformada y escalada, mientras que la interpretacion usa el CSV fisico con unidades y banderas de procedencia. Se distinguen explicitamente `observed`, `physically_derived` e `imputed`. Las derivaciones fisicas usadas incluyen $pl_dens = 5.514 \cdot pl_bmasse/pl_rade^3$ y $pl_orbsmax = (st_mass \cdot (pl_orbper/365.25)^2)^{1/3}$. `pl_dens` es mayoritariamente derivada desde `pl_bmasse` y `pl_rade`; por tanto, no debe tratarse como observacion independiente cuando se usa junto con masa y radio.

3 Mapper Methodology

Sea $X = \{x_1, \ldots, x_n\} \subset \mathbb{R}^p$. Sea $f : X \to \mathbb{R}^d$ un lens y $\mathcal{U} = \{U_\alpha\}$ una cubierta del espacio de lentes. Para cada U_α se clusteriza $f^{-1}(U_\alpha)$ y se construye un grafo donde los nodos son clusters locales y las aristas representan interseccion no vacia de miembros. Usamos $\beta_1 = E - V + C$ con E aristas, V nodos y $C = \beta_0$ componentes conexas. La configuracion base usa n_cubes=10, overlap=0.35, DBSCAN, min_samples=4, eps_percentile=90 y random_state=42.

4 Mapper Results

Los resultados principales se resumen en las figuras estaticas globales y en los grafos `joint_no_density` y `joint` con lens `pca2`. La configuracion con mayor complejidad combinada fue `thermal_density_cubes10_overlap` con 205 nodos y β_1=171. Esto no prueba clases planetarias finales: resume estructura inducida por una matriz completada y auditada.

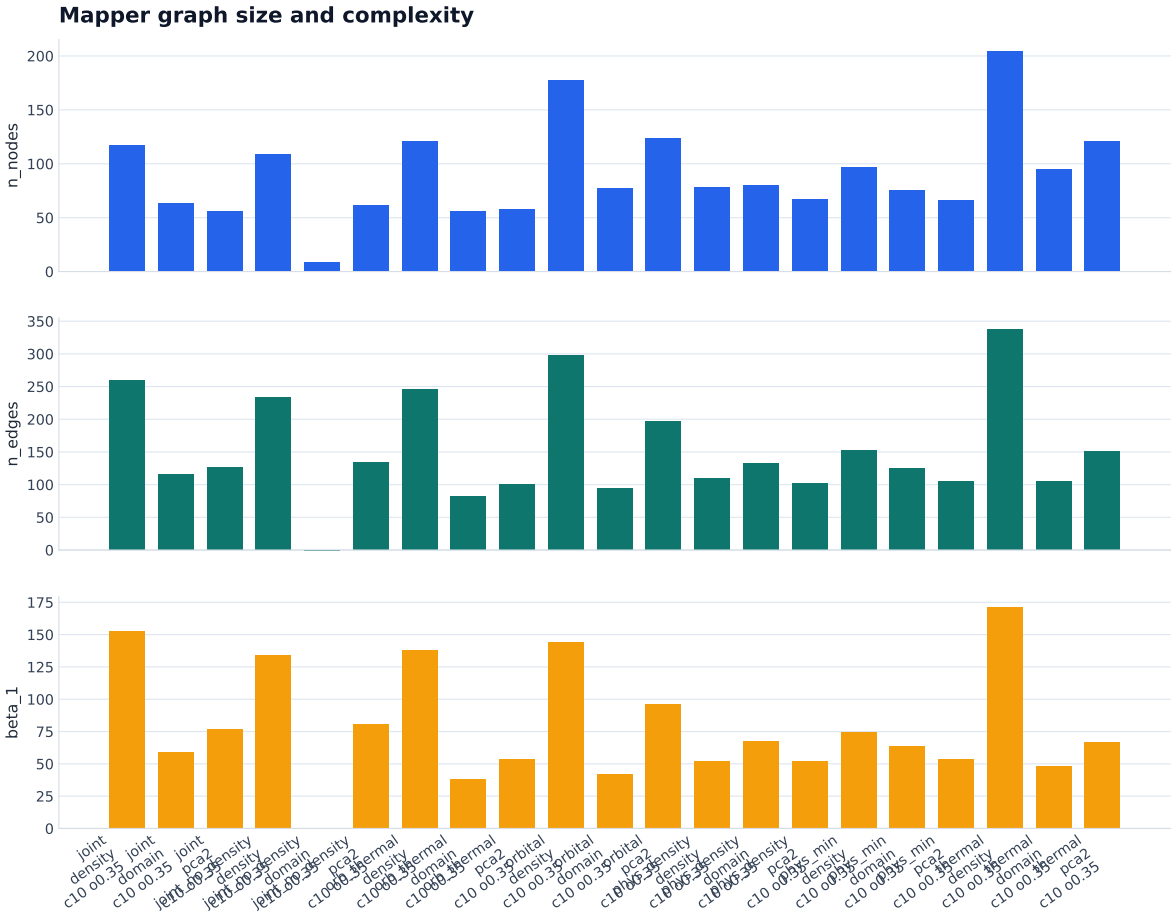

Figure 1: Tamano y complejidad de los grafos Mapper.

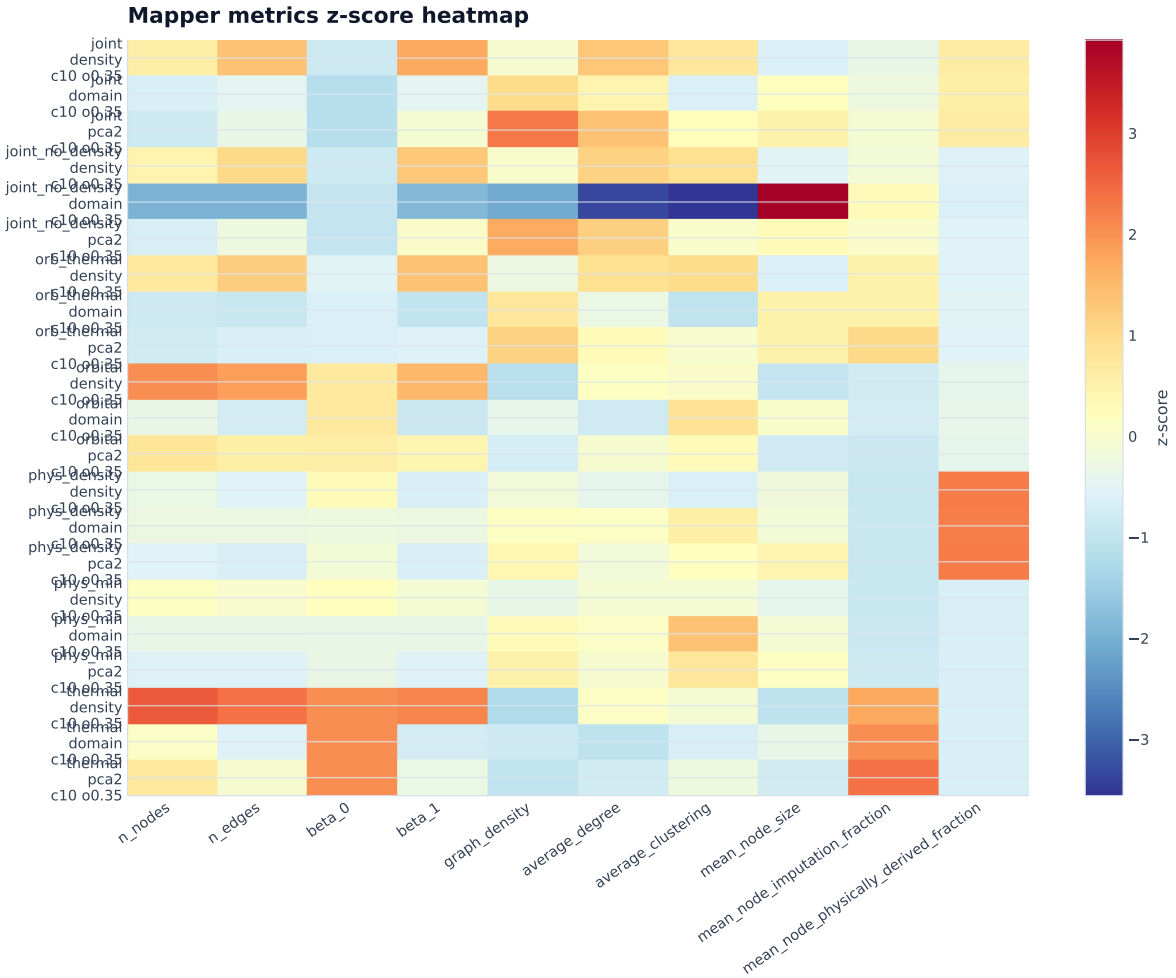

Figure 2: Heatmap de metricas estandarizadas.

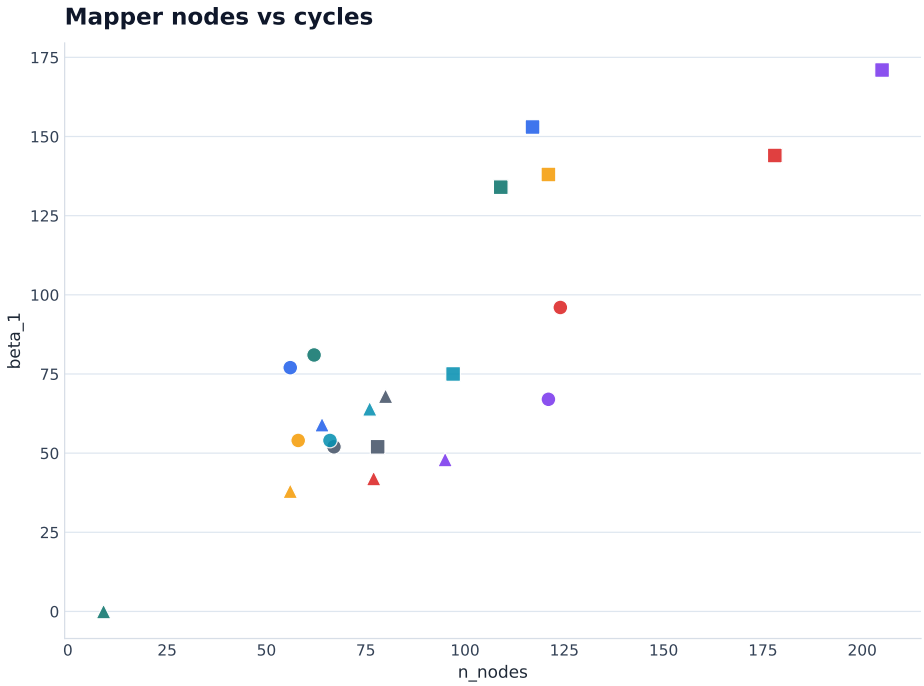

Figure 3: Relacion entre numero de nodos y ciclos.

5 Sensitivity Analysis

Adding `pl_dens` reduces the number of independent cycles, suggesting that the derived density feature may simplify or collapse part of the Mapper geometry. The Mapper graph is lens-sensitive, so conclusions should be treated as exploratory. This configuration contains a high average node-level imputation fraction; its topology should be interpreted cautiously.

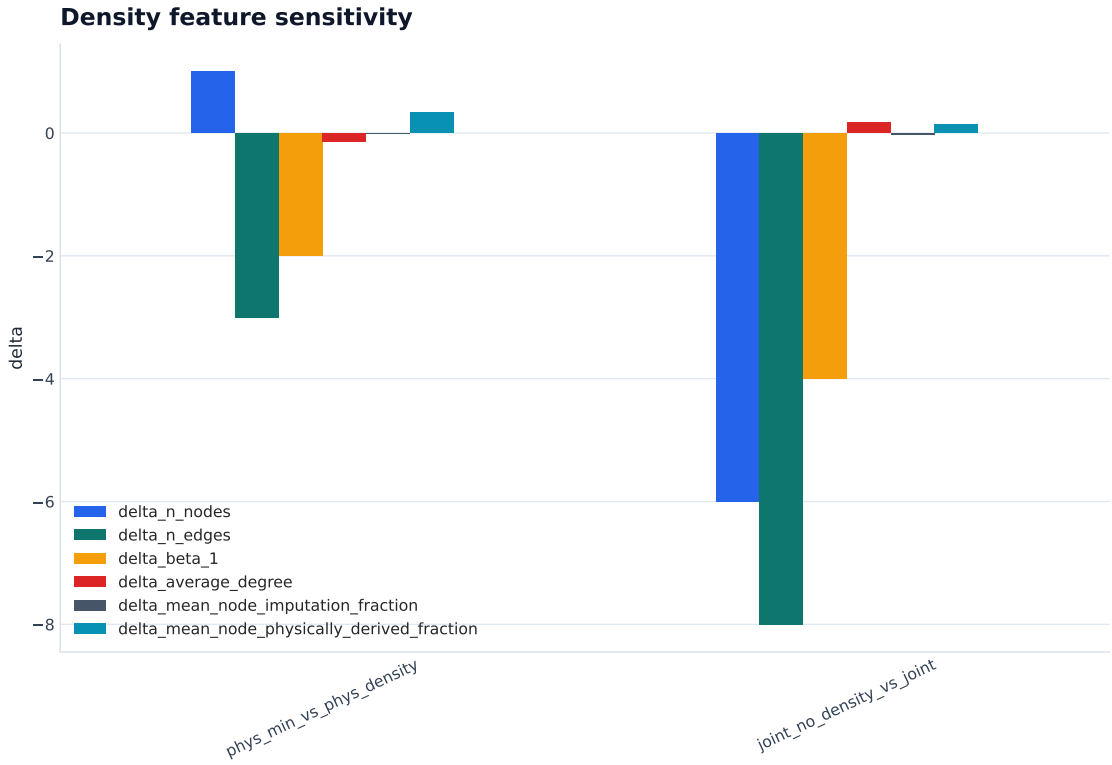

Figure 4: Sensibilidad al agregar `pl_dens`.

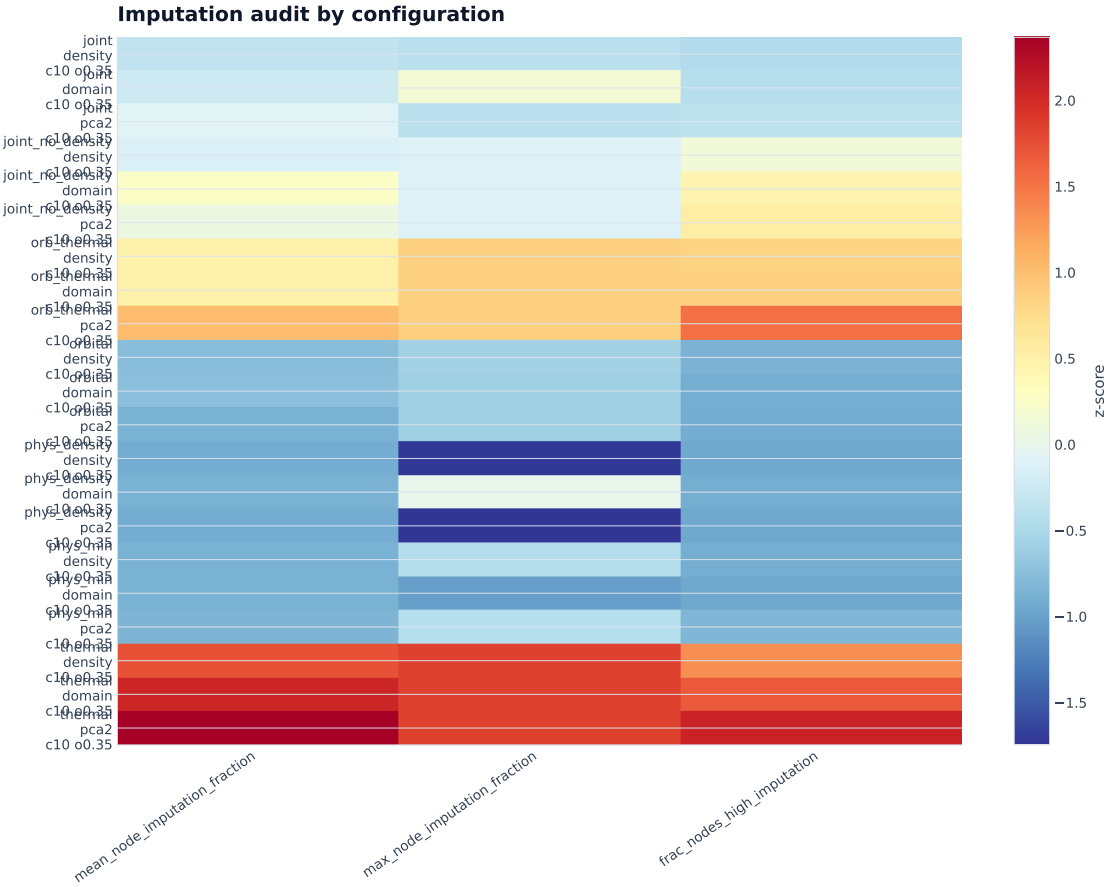

Figure 5: Auditoria de imputacion por configuracion.

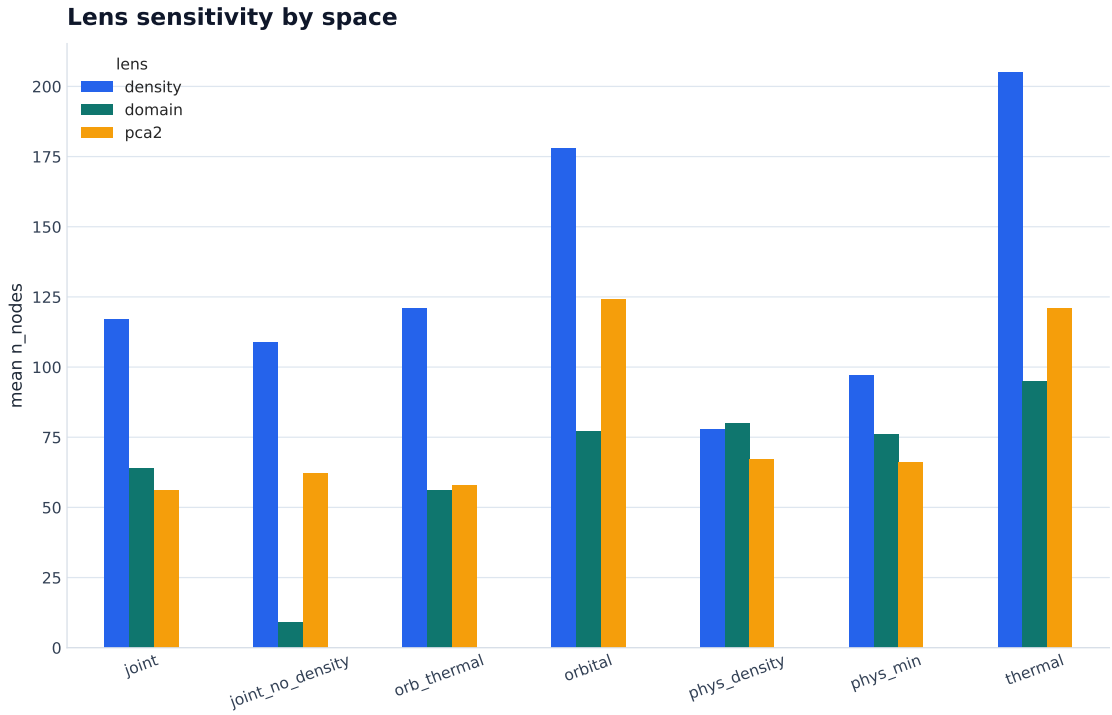

Figure 6: Sensibilidad al lens.

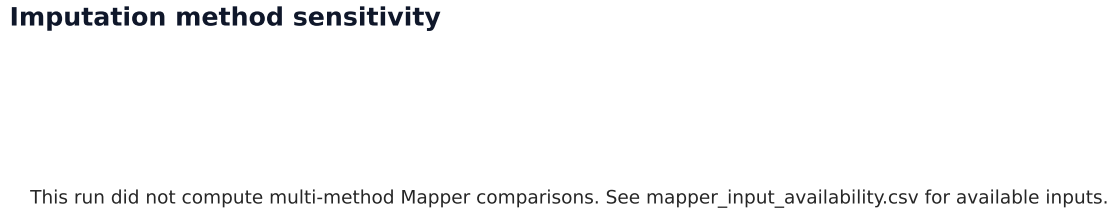

Figure 7: Sensibilidad al metodo de imputacion cuando hay insumos comparables.

6 Interpretation

Mapper muestra estructura topologica inducida por un espacio de variables completado y auditado. Las estructuras con mayor credibilidad son las que persisten entre espacios, lenses y controles

de imputacion, y que ademas tienen interpretacion fisica. Las regiones compatibles con rocosos, super-Tierras, sub-Neptunos, gigantes gaseosos o hot Jupiters solo deben enfatizarse si los nodos correspondientes muestran consistencia fisica y baja dependencia de imputacion.

7 Limitations

Las limitaciones principales incluyen sesgo observacional por `discoverymethod`, variables derivadas no independientes, imputacion que no equivale a observacion, dependencia de parametros Mapper, y el hecho de que un β_1 alto no implica automaticamente estructura fisica. La distancia `metric_zscore_l2` es una distancia entre vectores de metricas estandarizadas, no una distancia topologica estricta. `observed` $\neq$ `physically_derived` $\neq$ `imputed`.

8 Conclusion

El pipeline Mapper quedo actualizado para usar iterative por defecto, generar outputs estaticos en outputs/mapper y producir un reporte LaTeX autocontenido. Los siguientes pasos recomendados son estabilidad por bootstrap y revision cientifica de nodos destacados.

Table 1: Mapper graph metrics summary.

config_id	input_method	space	lens	n_cubes	overlap	clusterer	min_samples	eps_percentile	rows_used	n_features	mean_node_imputation_fraction	max_node_imputation_fraction	mean_node_physically_derived_fraction	frac_nodes_high_imputation	frac_nodes_high_derived	config_has_density_feature	n_nodes	n_edges	beta_0	beta_1	graph_density	average_degree	average_clustering	transitivity	n_isolates	largest_component_size	largest_component_fraction	mean_node_size	median_node_size	min_node_size	max_node_size
joint_density_cubes10_overlap0p35	iterative	joint	density	10	0.350	dbscan	4	90.000	6273	7	0.106	0.429	0.147	0.120	0.000	True	117	260	10	153	0.038	4.444	0.652	0.213	4	103	0.880	113.256	10.000	3	1685
joint_domain_cubes10_overlap0p35	iterative	joint	domain	10	0.350	dbscan	4	90.000	6273	7	0.121	0.571	0.139	0.125	0.000	True	64	116	7	59	0.058	3.625	0.445	0.333	2	54	0.844	209.797	8.000	3	2552
joint_pca2_cubes10_overlap0p35	iterative	joint	pca2	10	0.350	dbscan	4	90.000	6273	7	0.152	0.429	0.148	0.143	0.000	True	56	126	7	77	0.082	4.500	0.573	0.466	0	40	0.714	246.054	23.000	4	2384
joint_no_density_density_cubes10_overlap0p35	iterative	joint_no_density	density	10	0.350	dbscan	4	90.000	6273	6	0.142	0.500	0.005	0.257	0.000	False	169	253	10	134	0.040	4.275	0.675	0.206	4	93	0.853	122.101	7.000	3	1797
joint_no_density_domain_cubes10_overlap0p35	iterative	joint_no_density	domain	10	0.350	dbscan	4	90.000	6273	6	0.212	0.500	0.001	0.333	0.000	False	9	0	9	0	0.000	0.000	0.000	0.000	9	1	0.111	651.333	4.000	3	5825
joint_no_density_pca2_cubes10_overlap0p35	iterative	joint_no_density	pca2	10	0.350	dbscan	4	90.000	6273	6	0.176	0.500	0.008	0.355	0.000	False	62	134	9	81	0.071	4.323	0.542	0.454	3	47	0.758	220.645	23.000	3	2500
orb_thermal_density_cubes10_overlap0p35	iterative	orb_thermal	density	10	0.350	dbscan	4	90.000	6273	4	0.255	0.750	0.009	0.421	0.000	False	121	246	13	138	0.034	4.066	0.688	0.232	6	97	0.802	111.843	6.000	3	2520
orb_thermal_domain_cubes10_overlap0p35	iterative	orb_thermal	domain	10	0.350	dbscan	4	90.000	6273	4	0.254	0.750	0.011	0.429	0.000	False	56	82	12	38	0.053	2.929	0.384	0.355	6	40	0.714	242.607	8.000	3	2762
orb_thermal_pca2_cubes10_overlap0p35	iterative	orb_thermal	pca2	10	0.350	dbscan	4	90.000	6273	4	0.347	0.750	0.009	0.586	0.000	False	58	100	12	54	0.060	3.448	0.536	0.476	4	36	0.621	244.793	6.500	3	2847
orbital_density_cubes10_overlap0p35	iterative	orbital	density	10	0.350	dbscan	4	90.000	6273	2	0.030	0.385	0.028	0.022	0.000	False	178	297	25	144	0.019	3.337	0.543	0.135	6	117	0.657	76.090	7.000	3	2389
orbital_domain_cubes10_overlap0p35	iterative	orbital	domain	10	0.350	dbscan	4	90.000	6273	2	0.034	0.375	0.030	0.013	0.000	False	77	94	25	42	0.032	2.442	0.665	0.862	4	13	0.169	189.896	6.000	3	3206
orbital_pca2_cubes10_overlap0p35	iterative	orbital	pca2	10	0.350	dbscan	4	90.000	6273	2	0.011	0.375	0.028	0.008	0.000	False	124	196	24	96	0.026	3.161	0.577	0.215	4	62	0.500	91.573	6.000	3	3316

Table 2: Mapper space comparison.

config_id	input_method	space	lens	n_cubes	overlap	clusterer	min_samples	eps_percentile	rows_used	n_features	mean_node_imputation_fraction	max_node_imputation_fraction	mean_node_physically_derived_fraction	frac_nodes_high_imputation	frac_nodes_high_derived	config_has_density_feature	n_nodes	n_edges	beta_0	beta_1	graph_density	average_degree	average_clustering	transitivity	n_isolates	largest_component_size	largest_component_fraction	mean_node_size	median_node_size	min_node_size	max_node_size
phys_min_pca2_cubes10_overlap0p35	iterative	phys_min	pca2	10	0.350	dbscan	4	90.000	6273	2	0.017	0.421	0.000	0.030	0.000	False	66	105	15	54	0.049	3.182	0.649	0.579	6	33	0.500	202.545	7.000	3	2323
phys_density_pca2_cubes10_overlap0p35	iterative	phys_density	pca2	10	0.350	dbscan	4	90.000	6273	3	0.002	0.079	0.332	0.000	0.985	True	67	102	17	52	0.046	3.045	0.561	0.529	7	33	0.493	238.388	6.000	3	3023
orbital_pca2_cubes10_overlap0p35	iterative	orbital	pca2	10	0.350	dbscan	4	90.000	6273	2	0.011	0.375	0.028	0.008	0.000	False	124	196	24	96	0.026	3.161	0.577	0.215	4	62	0.500	91.573	6.000	3	3316
thermal_pca2_cubes10_overlap0p35	iterative	thermal	pca2	10	0.350	dbscan	4	90.000	6273	2	0.588	1.000	0.000	0.719	0.000	False	121	150	38	67	0.021	2.479	0.505	0.521	6	35	0.289	96.157	6.000	3	2995
orb_thermal_pca2_cubes10_overlap0p35	iterative	orb_thermal	pca2	10	0.350	dbscan	4	90.000	6273	4	0.347	0.750	0.009	0.586	0.000	False	58	100	12	54	0.060	3.448	0.536	0.476	4	36	0.621	244.793	6.500	3	2847
joint_no_density_pca2_cubes10_overlap0p35	iterative	joint_no_density	pca2	10	0.350	dbscan	4	90.000	6273	6	0.176	0.500	0.008	0.355	0.000	False	62	134	9	81	0.071	4.323	0.542	0.454	3	47	0.758	220.645	23.000	3	2500
joint_pca2_cubes10_overlap0p35	iterative	joint	pca2	10	0.350	dbscan	4	90.000	6273	7	0.152	0.429	0.148	0.143	0.000	True	56	126	7	77	0.082	4.500	0.573	0.466	0	40	0.714	246.054	23.000	4	2384

Table 3: Density feature sensitivity.

comparison	without_density	with_density	delta_n_nodes	delta_n_edges	delta_beta_1	delta_average_degree	delta_mean_node_imputation_fraction	delta_mean_node_physically_derived_fraction
phys_min_vs_phys_density	phys_min	phys_density	1.000	-3.000	-2.000	-0.137	-0.016	0.332
joint_no_density_vs_joint	joint_no_density	joint	-6.000	-8.000	-4.000	0.177	-0.024	0.140

Table 4: Lens sensitivity summary.

config_id	input_method	space	lens	n_cubes	overlap	clusterer	min_samples	eps_percentile	rows_used	n_features	mean_node_imputation_fraction	max_node_imputation_fraction	mean_node_physically_derived_fraction	frac_nodes_high_imputation	frac_nodes_high_derived	config_has_density_feature	n_nodes	n_edges	beta_0	beta_1	graph_density	average_degree	average_clustering	transitivity	n_isolates	largest_component_size	largest_component_fraction	mean_node_size	median_node_size	min_node_size	max_node_size
joint_density_cubes10_overlap0p35	iterative	joint	density	10	0.350	dbscan	4	90.000	6273	7	0.106	0.429	0.147	0.120	0.000	True	117	260	10	153	0.038	4.444	0.652	0.213	4	103	0.880	113.256	10.000	3	1685
joint_domain_cubes10_overlap0p35	iterative	joint	domain	10	0.350	dbscan	4	90.000	6273	7	0.121	0.571	0.139	0.125	0.000	True	64	116	7	59	0.058	3.625	0.445	0.333	2	54	0.844	209.797	8.000	3	2552
joint_pca2_cubes10_overlap0p35	iterative	joint	pca2	10	0.350	dbscan	4	90.000	6273	7	0.152	0.429	0.148	0.143	0.000	True	56	126	7	77	0.082	4.500	0.573	0.466	0	40	0.714	246.054	23.000	4	2384
joint_no_density_density_cubes10_overlap0p35	iterative	joint_no_density	density	10	0.350	dbscan	4	90.000	6273	6	0.142	0.500	0.005	0.257	0.000	False	169	253	10	134	0.040	4.275	0.675	0.206	4	93	0.853	122.101	7.000	3	1797
joint_no_density_domain_cubes10_overlap0p35	iterative	joint_no_density	domain	10	0.350	dbscan	4	90.000	6273	6	0.212	0.500	0.001	0.333	0.000	False	9	0	9	0	0.000	0.000	0.000	0.000	9	1	0.111	651.333	4.000	3	5825
joint_no_density_pca2_cubes10_overlap0p35	iterative	joint_no_density	pca2	10	0.350	dbscan	4	90.000	6273	6	0.176	0.500	0.008	0.355	0.000	False	62	134	9	81	0.071	4.323	0.542	0.454	3	47	0.758	220.645	23.000	3	2500
orb_thermal_density_cubes10_overlap0p35	iterative	orb_thermal	density	10	0.350	dbscan	4	90.000	6273	4	0.255	0.750	0.009	0.421	0.000	False	121	246	13	138	0.034	4.066	0.688	0.232	6	97	0.802	111.843	6.000	3	2520
orb_thermal_domain_cubes10_overlap0p35	iterative	orb_thermal	domain	10	0.350	dbscan	4	90.000	6273	4	0.254	0.750	0.011	0.429	0.000	False	56	82	12	38	0.053	2.929	0.384	0.355	6	40	0.714	242.607	8.000	3	2762
orb_thermal_pca2_cubes10_overlap0p35	iterative	orb_thermal	pca2	10	0.350	dbscan	4	90.000	6273	4	0.347	0.750	0.009	0.586	0.000	False	58	100	12	54	0.060	3.448	0.536	0.476	4	36	0.621	244.793	6.500	3	2847
orbital_density_cubes10_overlap0p35	iterative	orbital	density	10	0.350	dbscan	4	90.000	6273	2	0.030	0.385	0.028	0.022	0.000	False	178	297	25	144	0.019	3.337	0.543	0.135	6	117	0.657	76.090	7.000	3	2389
orbital_domain_cubes10_overlap0p35	iterative	orbital	domain	10	0.350	dbscan	4	90.000	6273	2	0.034	0.375	0.030	0.013	0.000	False	77	94	25	42	0.032	2.442	0.665	0.862	4	13	0.169	189.896	6.000	3	3206
orbital_pca2_cubes10_overlap0p35	iterative	orbital	pca2	10	0.350	dbscan	4	90.000	6273	2	0.011	0.375	0.028	0.008	0.000	False	124	196	24	96	0.026	3.161	0.577	0.215	4	62	0.500	91.573	6.000	3	3316
phys_density_density_cubes10_overlap0p35	iterative	phys_density	density	10	0.350	dbscan	4	90.000	6273	3	0.002	0.084	0.332	0.000	0.987	True	78	109	21	52	0.036	2.795	0.444	0.451	6	39	0.500	164.974	18.000	4	2431
phys_density_domain_cubes10_overlap0p35	iterative	phys_density	domain	10	0.350	dbscan	4	90.000	6273	3	0.010	0.527	0.328	0.013	0.975	True	80	132	16	68	0.042	3.300	0.622	0.482	4	44	0.550	172.525	6.000	4	2563
phys_density_pca2_cubes10_overlap0p35	iterative	phys_density	pca2	10	0.350	dbscan	4	90.000	6273	3	0.002	0.079	0.332	0.000	0.985	True	67	102	17	52	0.046	3.045	0.561	0.529	7	33	0.493	238.388	6.000	3	3023
phys_min_density_cubes10_overlap0p35	iterative	phys_min	density	10	0.350	dbscan	4	90.000	6273	2	0.010	0.417	0.000	0.010	0.000	False	97	152	20	75	0.033	3.134	0.521	0.430	4	53	0.546	138.000	9.000	3	1735
phys_min_domain_cubes10_overlap0p35	iterative	phys_min	domain	10	0.350	dbscan	4	90.000	6273	2	0.010	0.269	0.000	0.000	0.000	False	76	125	15	64	0.044	3.289	0.745	0.603	2	34	0.447	173.513	7.000	3	2380
phys_min_pca2_cubes10_overlap0p35	iterative	phys_min	pca2	10	0.350	dbscan	4	90.000	6273	2	0.017	0.421	0.000	0.030	0.000	False	66	105	15	54	0.049	3.182	0.649	0.579	6	33	0.500	202.545	7.000	3	2323

Table 5: Imputation audit summary.

config_id	mean_node_imputation_fraction	max_node_imputation_fraction	frac_nodes_high_imputation
joint_density_cubes10_overlap0p35	0.106	0.429	0.120
joint_domain_cubes10_overlap0p35	0.121	0.571	0.125
joint_pca2_cubes10_overlap0p35	0.152	0.429	0.143
joint_no_density_density_cubes10_overlap0p35	0.142	0.500	0.257
joint_no_density_domain_cubes10_overlap0p35	0.212	0.500	0.333
joint_no_density_pca2_cubes10_overlap0p35	0.176	0.500	0.355
orb_thermal_density_cubes10_overlap0p35	0.255	0.750	0.421
orb_thermal_domain_cubes10_overlap0p35	0.254	0.750	0.429
orb_thermal_pca2_cubes10_overlap0p35	0.347	0.750	0.586
orbital_density_cubes10_overlap0p35	0.030	0.385	0.022
orbital_domain_cubes10_overlap0p35	0.034	0.375	0.013
orbital_pca2_cubes10_overlap0p35	0.011	0.375	0.008
phys_density_density_cubes10_overlap0p35	0.002	0.084	0.000
phys_density_domain_cubes10_overlap0p35	0.010	0.527	0.013
phys_density_pca2_cubes10_overlap0p35	0.002	0.079	0.000
phys_min_density_cubes10_overlap0p35	0.010	0.417	0.010
phys_min_domain_cubes10_overlap0p35	0.010	0.269	0.000
phys_min_pca2_cubes10_overlap0p35	0.017	0.421	0.030
thermal_density_cubes10_overlap0p35	0.472	1.000	0.541
thermal_domain_cubes10_overlap0p35	0.527	1.000	0.621
thermal_pca2_cubes10_overlap0p35	0.588	1.000	0.719